- **Real-time Inference**: Real-time inference (a.k.a. model serving) refers to the process of exposing your models as REST API endpoints. This enables highly available, low-latency predictions and is especially useful when deploying models that generate predictions required by real-time applications.

All these options are available via the Databricks UI, shown below. Let's use the Favorita Store example again. Perhaps we've met with our beverages business team, and they would like to see weekly forecasts to help decide how much of each product they should purchase. We will opt for batch inference since we only need to produce an updated forecast once a week. It just takes a few clicks to set up a model for batch processing. Follow along in the "Applying our learning" section of the Favorita Sales dataset for detailed instructions on deploying your model for batch inference.

Model Serving

Let's dive deeper into real-time inference or **Model Serving**. The process of model serving can be complex and costly, involving additional tools and systems to achieve real-time needs. Fortunately, deploying a registered model as an endpoint only takes a single click from the Databricks platform! Because Model Serving is tightly integrated with MLflow, the path from development to production is much faster. Using a model registered in the MLflow Model Registry, Databricks automatically prepares a production-ready container and deploys the container to serverless compute.

It's also easy to deploy models via API, as shown below. A forecasting problem doesn't make much sense as a model serving use case, so let's consider instead the MIC problem we've been working on throughout the book. We will use model serving serve our classification model in real-time. The following example code snippet creates an endpoint that serves a version of a model named `asl_model`.

```
POST /api/2.0/serving-endpoints

{
  "name": "lakehouse-in-action-endpoint",
  "config":{
   "served_models": [{
     "model_name": "asl_model",
     "model_version": "1",
     "workload_size": "Small",
     "scale_to_zero_enabled": true
    }]
  }
}
```

Figure 7.4 – Deploying a model through a serving endpoint